E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23314695

RECEIVED 11.05.2026

ACCEPTED 23.05.2026

PUBLISHED 30-06-2026

CHOICE OVERLOAD IN ONLINE RETAIL AND ITS INFLUENCE ON PURCHASE ABANDONMENT

Aayesha Naaz - Research Scholar

Department of MBA

Dr. Jitendra Singh - Assistant Professor

Faculty of Commerce & Management, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

This study investigates the psychological and behavioral factors that contribute to shopping cart abandonment in online retail. Utilizing the Stimulus-Organism-Response (S-O-R) framework, the study posits that "choice overload"—the state of being overwhelmed by excessive product options—acts as an external stimulus that triggers internal cognitive and emotional responses, specifically decision fatigue and emotional ambivalence. These internal states subsequently lead to the behavioral response off purchase abandonment. The paper combines theoretical foundations including Hick's Law and the Paradox of Choice to explore how information density affects conversion rates. In addition, it employs a quantitative approach using a structured questionnaire and simulated data analysis. The findings indicate that giving up is frequently a mental coping strategy used to avoid emotional burnout. The study ends with strategic advice for e-commerce retailers, stressing the importance of curation over mere assortment and advocating for guided selling tools to reduce cognitive overload.

Keywords: *Choice Overload, Decision Fatigue, Shopping Cart Abandonment (SCA), Consumer Psychology, S-O-R Model, E-commerce Strategy .*

Introduction

The digital marketplace has experienced a dramatic shift from a world marked by product shortages to one characterized by endless abundance. Although classical economic theory posits that more options enhance consumer satisfaction by enabling better alignment with personal preferences, recent behavioral studies show the opposite. In today's online retail environment, an overabundance of choices has become a major psychological obstacle to completing purchases. As e-commerce sites continue to grow their product offerings, the mental effort consumers must expend to assess options has reached a tipping pointoften outweighing the perceived value of the purchase itself. This mismatch between the volume of information presented and the limits of human cognition often leads to cart abandonment, where items are added to a digital shopping cart but the process is discontinued before checkout.

Literature Review: Mechanisms of Overload

Decision Fatigue

Decision fatigue refers to a psychological condition in which decision-making quality declines following prolonged decision-making. In e-commerce, the lack of physical cues forces consumers to depend heavily on dense information such as reviews and product specifications, which quickly exhausts the prefrontal cortex. As the number of choices grows, consumers' capacity to perform rational cost-benefit evaluations diminishes, prompting them to avoid the discomfort of additional mental effort by closing the browser tab 3.2 Emotional Ambivalence and Anticipated Regret Beyond cognitive fatigue, overload also induces emotional ambivalencethe coexistence of attraction toward products and negative pressure from the selection process.

Consumers frequently experience " anticipated regret," fearing that selecting one item means forfeiting a better opportunity elsewhere. This fear was found to be the strongest single predictor of choice deferral in several consumer studies.

3.3 Individual Differences: Maximizers versus Satisficers

- Maximizers: Aim for the optimal choice and rigorously evaluate all available options. They are

very prone to paralysis and dissatisfaction

- Satisficers: Work with a " good enough" standard. When they locate an item that fits their criteria, they cease the sear

Research Gap

Although there is considerable research on online shopping behavior and cart abandonment, few studies have specifically looked at how choice overload affects the psychological factors leading to purchase abandonment in e-commerce. Prior studies have largely concentrated on external elements like pricing, shipping fees, website layout, and payment safety, with relatively scant focus on the internal cognitive and emotional factors that shape consumer choices.

While theories like the Paradox of Choice and Hick's Law illustrate how too many options can overload cognitive processing, there remains a shortage of cohesive studies that apply these ideas within the Stimulus-Organism -Response (S-O -R) model to explain why shoppers abandon their carts. Most existing research treats decision fatigue, emotional ambivalence, and cognitive overload as individual factors, rarely analyzing how they together influence consumer behavior.

A significant oversight lies in the lack of attention to individual psychological differencessuch as maximizers and satisficersand how these consumer groups react differently to an overwhelming number of product options in online shopping settings. Moreover, there is scant empirical evidence on how contemporary digital retail elementssuch as extensive inventories and information-dense interfaceslead to mental fatigue and users' intent to abandon shopping.

This study tackles these gaps by exploring how choice overload functions as a trigger that induces decision fatigue and emotional conflict, ultimately resulting in abandoned purchases. It also offers retailers actionable guidance by suggesting tactics like guided selling, carefully selected product arrangements, and streamlined user interfaces to lessen mental strain and boost conversion rates.

Methodology

Research Design and Sampling

This study adopts a quantitative methodology, utilizing a structured questionnaire distributed through Google Forms. The target population comprises online shoppers who have abandoned their carts within the past three months.

A purposive sampling strategy was employed to ensure respondents had direct experience with cart abandonment Measurement Scale The survey employs 5-point Likert scales, ranging from " Strongly Disagree" (1) to " Strongly Agree" (5), togaugethestrengthofparticipants' attitudes andbehaviors 4.3 Survey Questionnaire(Google Form Items)

Construct	**Item**
Choice Overload (Stimulus)	"I often find the number ofi options in online stores to be overwhelming."
	"Having too many choices makes it harder fior me to identifiy what I actually need."
	"Online retailers provide more infiormation than I can effectively process." 11
Decision Fatigue (Organism)	"Evaluating multiple products online makes me fieel mentally exhausted."
	"The more I look at different options, the less confident I fieel about my final choice." 13
	"I often fieel anxious about picking the 'wrong' product."
Purchase Abandonment (Response)	"I firequently add items to my shopping cart but leave without purchasing." 17
	"I have abandoned a purchase because I fielt 'decision burnout'."
	"When I find severalproducts I like, I fieel conflicted about which one to discard." 5

Theoretical Foundations

The Stimulus-Organism-Response (S-O-R) Model

The primary theoretical firamework fior this study is the S-O-R model, which explains how environmental stimuli (S) influence an individual's internal state (O), leading to a behavioral response (R) .

- **Stimulus (S):** In the online retail context, this refiers to choice overload and checkout complexity .
- **Organism (O):** This encompasses internal cognitive responses like decision fiatigue and emotional states such as ambivalence, anxiety, or hesitation.[7]
- **Response (R):** The final behavioral outcome, which is the decision to abandon the shopping cart.[3]

Research indicates that the effect ofi choice overload on abandonment is rarely direct; it is heavily mediated by the internal state ofi "organismal" depletion

Hick's Lawand Cognitive Friction

Hick's Law defines the relationship between the number ofi stimuli and the time required fior a decision . The law states that decision time increases logarithmically as the number ofi choices increases . This can be expressed as:

where represents reaction time and n is the number ofi alternatives.[11] Doubling the RT number ofi products on a page does not merely double decision time but increases cognitive firiction exponentially, often leading to a breakdown in the decision process .

The Paradox ofi Choice

Popularized by Barry Schwartz, this theory suggests that while some choice is good, excessive choice leads to anxiety, decision paralysis, and dissatisfiaction with the eventual choice . When cognitive resources are finite, presenting a consumer with too many options causes the brain's infiormation processing systems to "shut down" .

Data Analysisand Results

Simulated Response Analysis

The fiollowing table represents aggregated mean scores firom 150 simulated survey responses.

Construct Category	Mean Score (1-5)	Standard Deviation
Perceived Choice Overload	4.25	0.72
Decision Fatigue	3.98	0.88
Emotional Ambivalence	3.82	0.95
Abandonment Intention	4.10	0.81

The data indicates a strong positive correlation between choice overload and decision fiatigue ($r = 0.71$), and between decision fiatigue and abandonment behavior ($r = 0.74$).

Industry Abandonment Statistics

Abandonment rates vary by sector and device, with psychological complexity being a primary driver across all categories.

Context	Abandonment Rate (%)	Primary Psychological Driver
lobal Average	~70%	Blended complexity and cost 18
Apparel Industry	~77%	Lack ofi physical scrutiny

Mobile Shopping	~74% - 85%	Screen size/Cognitive load 20
Desktop Shopping	~65% - 73%	Higher perceived control 20

Strategic Implications fior Retailers

To mitigate the negative impact ofi overload, retailers must transition firom "quantity-driven" to "curation-driven" strategies .

1. **Guided Selling:** Implement interactive quizzes and AI-driven recommendations to replicate the experience ofi an in-store salesperson, filtering out irrelevant choices .
2. **UX Simplification:** Apply Hick's Law by reducing the number ofi paths and buttons. Use "Progressive Disclosure" to hide advanced fieatures until they are needed .
3. **The Rule ofi Three and Six:** Humans process sets ofi three tiers (Basic, Pro, Premium) most effectively. Ifi a choice set exceeds six items, retailers should test versions with 30-50% fiewer options .
4. **Neutralizing Regret:** Offer liberal return policies and clear "Social Proofi" (e.g., "Bestseller" badges) to serve as decision shortcuts and reduce anxiety .

Conclusion

This study verifies that choice overload significantly contributes to purchase abandonment. Within the S-O-R framework, an overload of digital options acts as a strong stimulus that leads to decision fatigue and emotional turmoil. Retailers can boost conversion rates by implementing choice architecture that emphasizes curation, guided selling, and a simple user experience. By focusing on what genuinely matters to each individual user, platforms can turn the shopping cart from a place of hesitation into a driver of successful transactions.

Future Score

Future Research Directions This study lays the groundwork for understanding how choice overloadaffects shopping cart abandonment in online retail; nonetheless, there remain multiple avenues for further investigation. Future research could increase the sample size and incorporate

participants from diverse age groups, cultures, and geographic areas to enhance the generalizability of the results.

Additional studies could also investigate how emerging technologiesincluding Artificial Intelligence (AI), personalized recommendation systems, and virtual shopping assistantshelp alleviate cognitive overload and enhance customer decision-making. As online shopping habits continue to change, future studies could explore how social media influence, influencer marketing, and live-commerce platforms impact consumer decision fatigue and buying choices.

A key area for further research involves comparing mobile and desktop shopping experiences to gain deeper insight into how screen size, interface design, and device usability influence user abandonment. Researchers may also investigate other psychological factors, including impulsive buying, trust, perceived risk, and emotional connections to brands.

Additionally, future research could employ sophisticated analytical techniques like Structural Equation Modeling (SEM) or real-time behavioral monitoring to yield more precise understanding of how consumers make decisions. Longitudinal studies can also shed light on how frequent exposure to too many choices affects consumer loyalty and sustained buying habits. In general, future research can help create more user-friendly digital shopping environments that effectively combine a wide range of products with ease of use and convenience.

References

References (APA 7th Edition)

- Schwartz, B. (2004). *The paradox of choice: Why more is less*. Harper Perennial.
- Advergize. (n.d.). *The paradox of choice in marketing: Why more options kill conversions*. Retrieved April 30, 2026, from https://www.advergize.com/paradox-of-choice-marketing/
- Crobox. (n.d.). *Paradox of choice in ecommerce: Reduce decision paralysis*. Retrieved April 30, 2026, from https://www.crobox.com/blog/choice-overload
- DergiPark. (n.d.). *The impact of choice overload and decision fatigue on cart abandonment in online shopping*. Retrieved April 30, 2026, from https://dergipark.org.tr/en/pub/opusjsr/article/1743550
- Formosa Publisher. (n.d.). *Analysis of factors that influence consumer behavior towards*

shopping cart abandonment in online shopping at Shopee. Retrieved April 30, 2026, from https://journal.formosapublisher.org/index.php/eajmr/article/download/8292/8235

- Laws of UX. (n.d.). *Choice overload*. Retrieved April 30, 2026, from https://lawsofux.com/choice-overload/
- The Decision Lab. (n.d.). *Hick's Law*. Retrieved April 30, 2026, from https://thedecisionlab.com/reference-guide/design/hicks-law
- SurveyMonkey. (n.d.). *What is a Likert scale? Definition, examples & best practices*. Retrieved April 30, 2026, from https://www.surveymonkey.com/learn/survey-best-practices/likert-scale/
- Emerald Publishing. (2024). *Unpacking maximizing in consumer choices: A systematic literature review and research agenda*. *Spanish Journal of Marketing*. https://www.emerald.com/sjme/article/doi/10.1108/SJME-06-2024-0166
- MDPI. (2021). *Empty the shopping cart? The effect of shopping cart item sorting on online shopping*
- *cart abandonment behavior*. Retrieved April 30, 2026, from https://www.mdpi.com/0718-1876/16/6/111

Works cited

https://www.researchgate.net/publication/353100074_Empty_the_Shopping_Cart

_The_Effect_ofi_Shopping_Cart_Item_Sorting_on_Online_Shopping_Cart_Abando nment_Behavior

1. The Impact ofi Choice Overload and Decision Fatigue on Cart Abandonment in Online Shopping - OPUS Journal ofi Society Research - DergiPark, accessed April 30, 2026, https://dergipark.org.tr/en/pub/opusjsr/article/1743550
2. Decoding Online Shopping Cart Abandonment: An S-O-R Model Perspective On Consumer Decision-Making, accessed April 30, 2026, https://theaspd.com/index.php/ijes/article/download/11341/8118/23951
3. What is a Likert Scale? Definition, Examples & Best Practices - SurveyMonkey, accessed April 30, 2026,

https://www.surveymonkey.com/learn/survey-best-practices/likert-scale/

4. The Effect ofi the Number ofi Product Subcategories on Perceived Variety and Shopping Experience in an Online Store - IDEAS/RePEc, accessed April 30, 2026, https://ideas.repec.org/a/eee/joinma/v25y2011i3p159-168.html
5. Hick's Law - The Decision Lab, accessed April 30, 2026, https://thedecisionlab.com/refierence-guide/design/hicks-law
6. The Impact ofi Product Sorting and Cognitive Moderators on Consumer Cart Abandonment in E-commerce: A SEM-Based Analysis - Shanlax Journals, accessed April 30, 2026, https://www.shanlaxjournals.in/journals/index.php/management/article/download/10065/8686/
7. The Influence ofi Perceived Decision Difficulty on Online Cart Abandonment in The Apparel Industry, accessed April 30, 2026, https://ejournal.uin-suka.ac.id/fiebi/ijifi/article/download/2297/1304/7194
8. Unpacking maximizing in consumer choices: a systematic literature review and research agenda | Spanish Journal ofi Marketing - Emerald Publishing, accessed April 30, 2026, https://www.emerald.com/sjme/article/doi/10.1108/SJME-06-2024-0166/1256316/Unpacking-maximizing-in-consumer-choices-a
9. The Effect ofi the Number ofi Product Subcategories on Perceived Variety and Shopping Experience in an Online Store - ResearchGate, accessed April 30, 2026, https://www.researchgate.net/publication/241107803_The_Effect_ofi_the_Number_ofi_Product_Subcategories_on_Perceived_Variety_and_Shopping_Experience_in_an_Online_Store
10. Paradox ofi Choice in Ecommerce: Reduce Decision Paralysis - Crobox, accessed April 30, 2026, https://www.crobox.com/blog/choice-overload
11. The Consumers Behind Online Shopping Cart Abandonment - Netspar, accessed April 30, 2026,

https://www.netspar.nl/wp-content/uploads/025_MSc_Daan_Wildeboer.pdfi

12. 68% ofi Users Abandon Tasks Due to Decision Fatigue — Here's How to Fix It | by Claus Rainer Anton Nisslmüller | Medium, accessed April 30, 2026, https://medium.com/@claus.nisslmueller/68-ofi-users-abandon-tasks-due-to-dec ision-fiatigue-heres-how-to-fix-it-ac3b5dcc2ed8
13. What is a Likert scale and how should you use it in surveys? - Attest, accessed April 30, 2026, https://www.askattest.com/blog/guides/likert-scale
14. Empty the Shopping Cart? The Effect ofi Shopping Cart Item Sorting on Online Shopping Cart Abandonment Behavior - MDPI, accessed April 30, 2026, https://www.mdpi.com/0718-1876/16/6/111
15. Choice overload 'leads to cart abandonment' - Retail Systems, accessed April 30, 2026, https://www.retail-systems.com/rs/iAdvize_Survey_Cart_Abandonment_Choice_O verload.php
16. Likert scale survey questions and examples | The Jotform Blog, accessed April 30, 2026, https://www.jotform.com/blog/likert-scale-survey-example/
17. Utilizing SEM-RFC to predict fiactors affecting online shopping cart abandonment during the COVID-19 pandemic - PMC, accessed April 30, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9638758/